

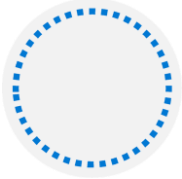
Training Course

Module 03: Azure Resources

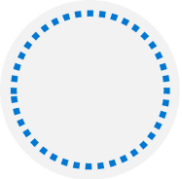




Lesson 01: Configure Azure Resources with Tools



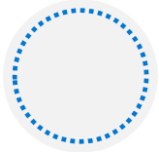
Lesson 02: Use Azure Resource Manager



Lesson 03: Configure Resources with ARM Templates

Lesson 01: Configure Azure Resources with Tools

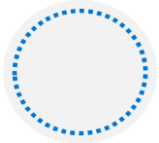




Use the Azure Portal



Use Azure Cloud Shell



Use Azure PowerShell



Use Azure CLI

Use the Azure Portal

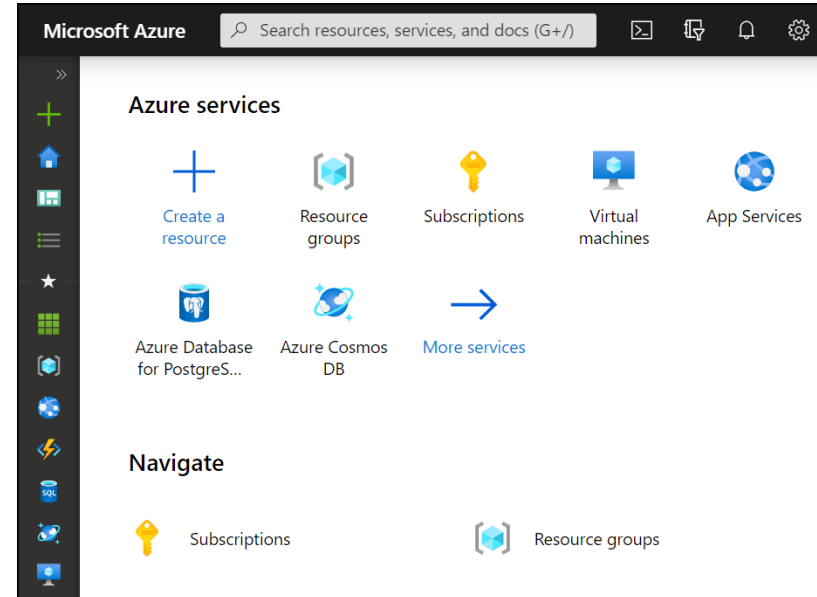
Search resources, services, and docs

Manage resources

Create customized dashboards and favorites

Access the Cloud Shell

Receive notifications



Use Azure Cloud Shell

Interactive, browser-accessible shell

Offers either Bash or PowerShell

Is temporary and provided on a per-session, per-user basis

Requires a resource group, storage account, and Azure File share

Authenticates automatically

Integrated graphical text editor

Is assigned one machine per user account

Times out after 20 minutes



Welcome to Azure Cloud Shell

Bash

PowerShell

```
New-AzVm `
  -ResourceGroupName "CrmTestingResourceGroup" `
  -Name "CrmUnitTests" `
  -Image "UbuntuLTS" `
...

```

- Connect to your Azure subscription and manage resources
- Adds the Azure-specific commands
- Available inside a browser via the Azure Cloud Shell
- Available as a local installation on Linux, macOS, or Windows
- Has an interactive and a scripting mode

```
az vm restart -g MyResourceGroup -n MyVm
```

- Cross-platform command-line program
- Runs on Linux, macOS, and Windows
- Can be used interactively or through scripts
- Commands are structured in *_groups_* and *_subgroups_*
- Use *find* to locate commands
- Use *--help* for more detailed information

Lesson 02: Use Azure Resource Manager





Review Resource Manager Benefits



Review Azure Resource Terminology



Create Resource Groups



Create Resource Manager Locks



Reorganize Azure Resources



Remove Resources and Resource Groups



Determine Resource Limits

Review Resource Manager Benefits

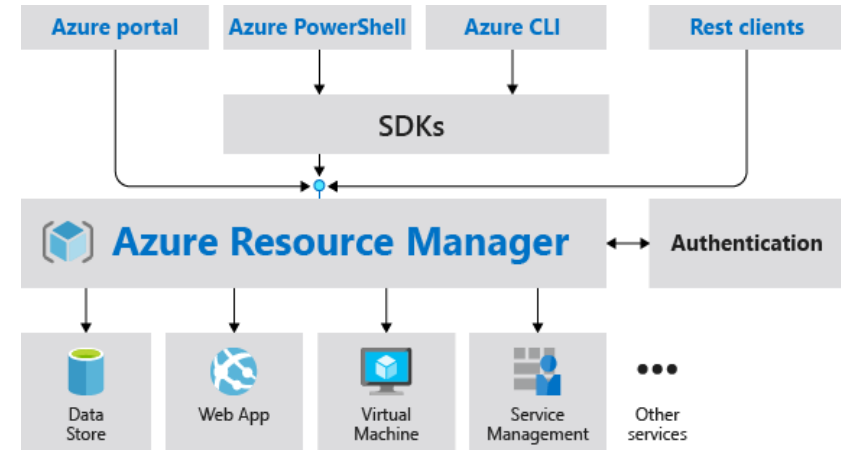
Provides a consistent management layer

Enables you to work with the resources in your solution as a group

Deploy, update, or delete in a single, coordinated operation

Provides security, auditing, and tagging features

Choose the tools and APIs that work best for you



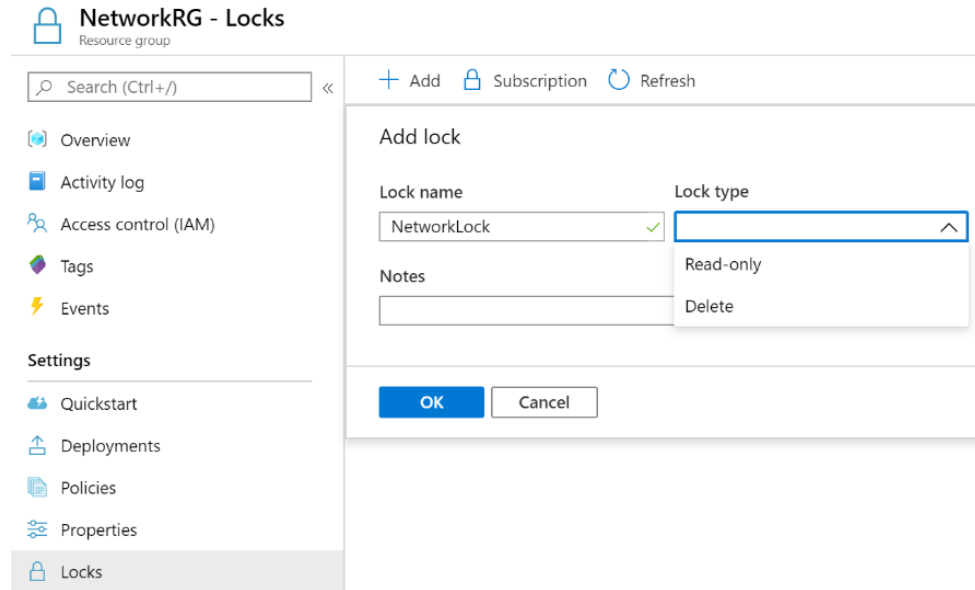
Create Resource Manager Locks

Associate the lock with a subscription, resource group, or resource

Locks are inherited by child resources

Read-Only locks prevent any changes to the resource

Delete locks prevent deletion



NetworkRG - Locks
Resource group

Search (Ctrl+/) <<

+ Add Subscription Refresh

Add lock

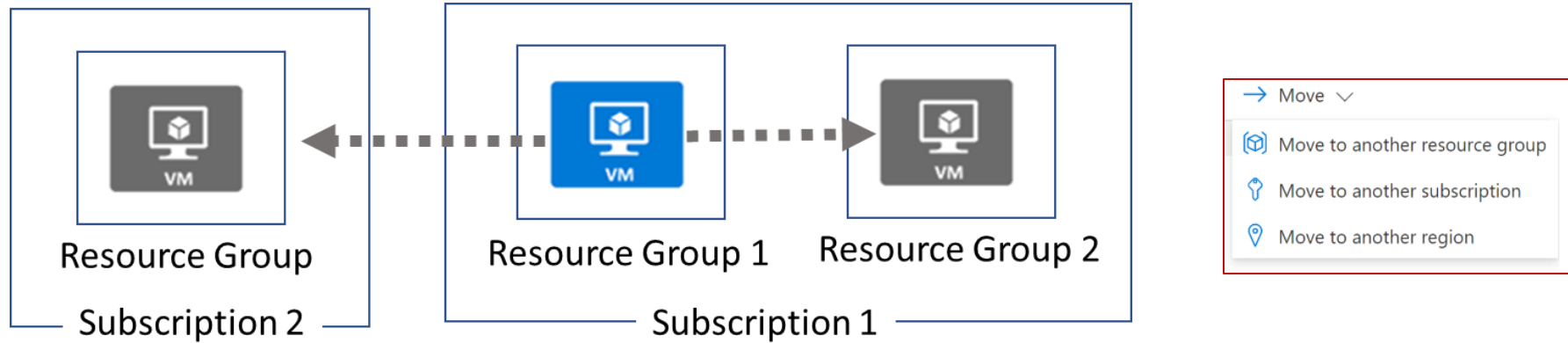
Lock name: NetworkLock ✓

Lock type: Read-only (selected from dropdown: Read-only, Delete)

Notes:

OK Cancel

Reorganize Azure Resources



When moving resources, both the source group and the target group are locked during the operation

Services that cannot be moved: Azure AD Domain Services, ExpressRoute, and Site Recovery. Other restrictions apply

Remove Resources and Resource Groups

Resource groups
(web + DB, VM, Storage) in one group



Refresh → Move **Delete**

Resource group (change) : ASH

Location : West Central US

Subscription (change) : Visual Studio Enterprise

`Get-AzResourceGroup -Name 'az104-03*' | Remove-AzResourceGroup -Force -AsJob`

Remove Azure resources
that you no longer use

Ensures you will not see
unexpected charges

Remove individual
resources or remove the
resource group

Determine Resource Limits

ASC DEMO | Usage + quotas

Subscription

Settings

Programmatic deployment

Resource groups

Resources

Usage + quotas

Policies

Security

Events

You can use each Microsoft Azure resource up to its quota. Each subscription has separate quotas and usage is tracked per subscription. If you reach a quota cap, you can request an increase via Help + Support. [Learn more](#)

Request Increase

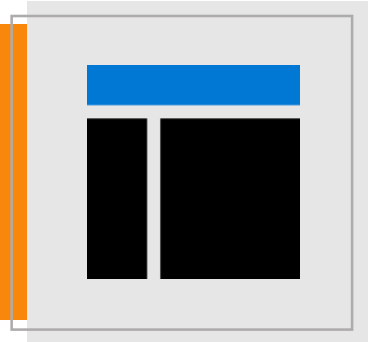
Quota	Provider	Location	Usage		
Total Regional vCPUs	Microsoft.Compute	East US		25 %	25 of 100
Total Regional vCPUs	Microsoft.Compute	West Europe		21 %	21 of 100
Total Regional vCPUs	Microsoft.Compute	Central US		17 %	17 of 100
Standard Dv2 Family vCPUs	Microsoft.Compute	West Europe		16 %	16 of 100
Standard Dsv2 Family vCPUs	Microsoft.Compute	Central US		14 %	14 of 100

Resources have a default limit also known as quota

Helpful to track current usage, and plan for future use

You can open a free support case to increase limits to published maximums

Lesson 03: Configure Resources with ARM Templates



Review ARM Template Advantages

Improves consistency

Express complex deployments

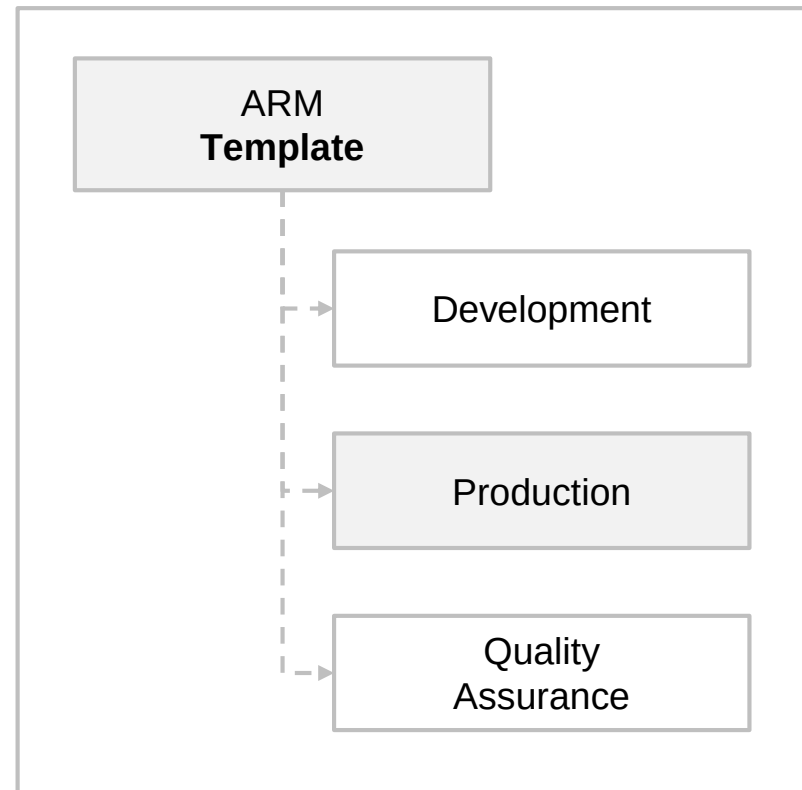
Reduce manual, error prone tasks

Express requirements through code

Promotes reuse

Modular and can be linked

Simplifies orchestration



Explore the ARM Template Schema

Defines all the Resource manager resources in a deployment

Written in JSON

A collection of key-value pairs

Each key is a string

Each value can be a string, number, Boolean expression, list of values, object

```
{
  "$schema":
    "http://schema.management.
    azure.com/schemas/2019-04-
    01/deploymentTemplate.json#",
  "contentVersion": "",
  "parameters": {},
  "variables": {},
  "functions": [],
  "resources": [],
  "outputs": {}
}
```

Explore the ARM Template Parameters

Specify which values are configurable when the template runs

This example has two parameters: one for a VM's username (adminUsername), and one for its password (adminPassword)

```
"parameters": {  
  "adminUsername": {  
    "type": "string",  
    "metadata": {  
      "description": "Username for the VM."  
    }  
  },  
  "adminPassword": {  
    "type": "securestring",  
    "metadata": {  
      "description": "Password for the VM."  
    }  
  }  
}
```

Review QuickStart Templates

Resource Manager templates provided
by the Azure community

Provides everything you need to deploy
your solution or serves as a starting
point for your template

757 Quickstart templates are currently in the gallery.

Create Configuration Manager Tech Preview Lab in Azure

This template creates a new System Center Configuration Manager Technical Preview Lab environment. It creates 4 new Azure VMs, configuring a new AD Domain Contr...



by [Yizhong Wu](#),
Last updated: 12/10/2018

Create a Standard Storage Account

This template creates a Standard Storage Account



by [Brian Moore](#),
Last updated: 12/4/2018

Deploy a Django app

This template uses the Azure Linux CustomScript extension to deploy an application. This example creates an Ubuntu VM, does a silent install of Python, Django...



by [Madhan Arumugam Ramakrishnan](#),
Last updated: 7/19/2018

Create a new AD Domain with 2 Domain Controllers

This template creates 2 new VMs to be AD DCs (primary and backup) for a new Forest and Domain



by [Simon Davies](#),
Last updated: 7/5/2018

Thank you

